

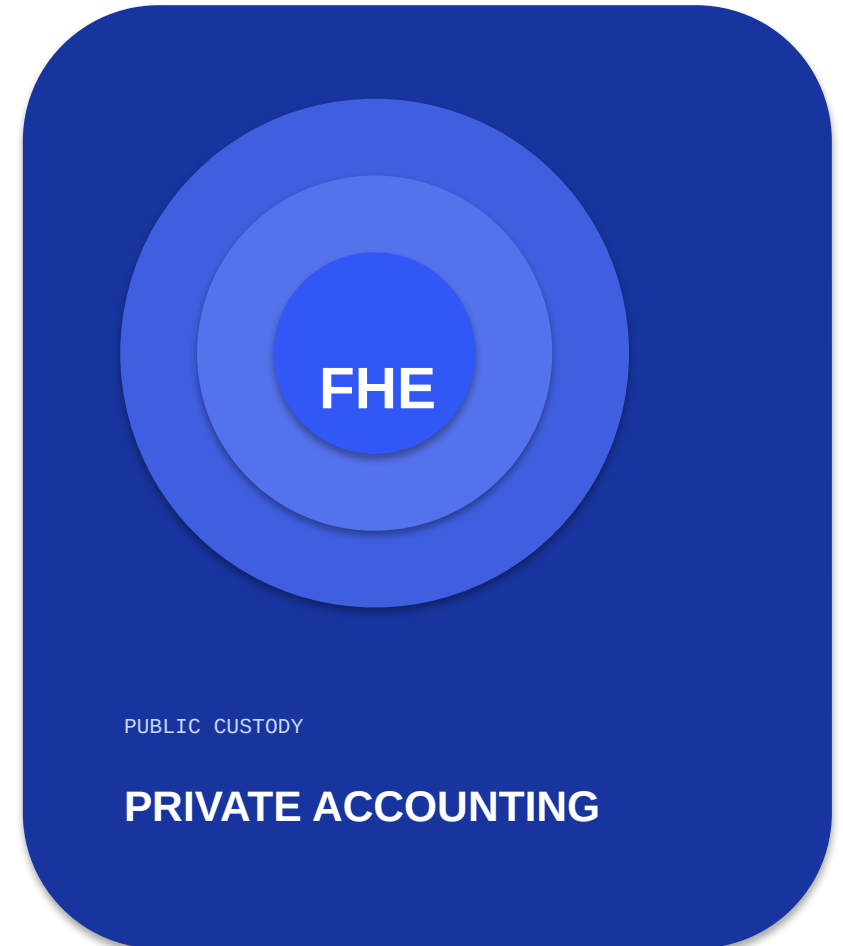
ZAMA FHEVM · CONFIDENTIAL FINANCE

Private prize savings, built for verification.

Principal stays withdrawable while encrypted balances and ticket weights stay confidential.

cipherpool-beta.vercel.app

RESEARCH SOFTWARE · NOT PRODUCTION



THE PROBLEM

Public ledgers expose private financial context

Auditability should not require publishing every saver's position.

01

Balance exposure

Wallet history can reveal a saver's position, deposits, withdrawals, and changing financial behavior.

02

Visible ticket weights

Transparent weights make whale activity easy to track and can disclose each participant's relative odds.

03

Timing reveals intent

Pending deposits and last-minute entries expose behavior around a prize round before it settles.

THE SOLUTION

Public custody. Confidential accounting.

CipherPool separates what must be auditable from what should remain private.

PUBLIC · INDEPENDENTLY VERIFIABLE

Custody layer

- Deposited testnet USDC
- Contract and vault bindings
- Transaction receipts
- Aggregate solvency

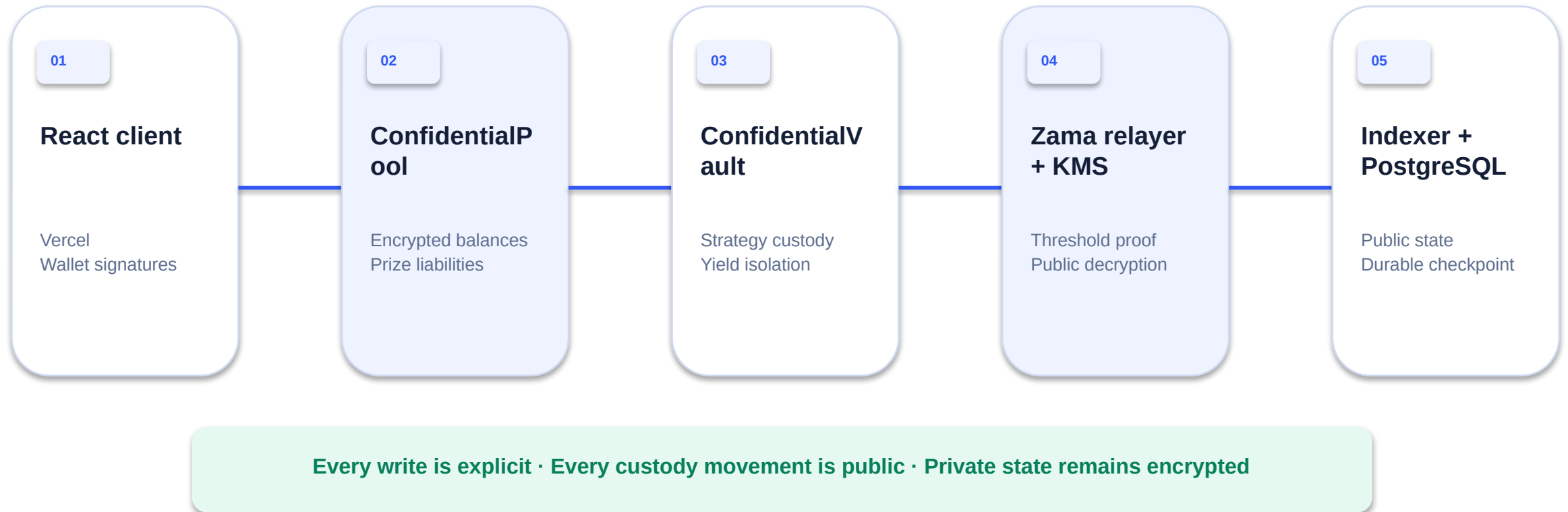
ENCRYPTED · WALLET-AUTHORIZED

Privacy layer

- Running balances as uint64
- Weighted draw intervals
- Prize credits
- Local balance reveal

ARCHITECTURE

A verifiable path from wallet to confidential settlement



LIVE PRODUCT

The console is live—and honest about connection state

The screenshot shows the CipherPool Confidential Savings Console. At the top, there's a header with the CipherPool logo, navigation links (Save privately, Prize rounds, How it works), and a Connect wallet button. The main content area is titled 'CONFIDENTIAL SAVINGS CONSOLE' and features a large heading 'Your position stays private.' followed by a description: 'Deposit testnet USDC, enter verifiable prize rounds, and withdraw through threshold decryption—without publishing your balance or ticket weight.' Below this are three icons: Encrypted accounting, Wallet-bound access, and Verifiable exits. A yellow warning banner states: 'Writes remain locked until runtime verification succeeds. Connect a Sepolia wallet to verify the active deployment.' Below the banner is a status bar showing 'Indexer Checking', 'Pool Unpaused', 'Ethereum Sepolia', and 'Deployment Pending'. The main dashboard is divided into four sections: 'ACCOUNTED BALANCES' (0 USDC, Principal + awarded prizes), 'AVAILABLE YIELD' (0 USDC, After reserved liabilities), 'PRIVATE SAVERS' (0, On-chain participants), and 'CONFIRMED ROUNDS' (0, Recorded on Sepolia). A 'LIVE' button is next to the Confirmed Rounds section. Below the dashboard is a 'PRIVATE OPERATIONS' section titled 'Manage your savings'. It contains three steps: 'STEP 01 Fund your private position' (with an 'ENCRYPTED BALANCE' tag), 'STEP 02 Withdraw without exposing your balance' (with a 'READY' tag), and a third step partially visible. Each step has a brief description of the process.

PRODUCTION CAPTURE

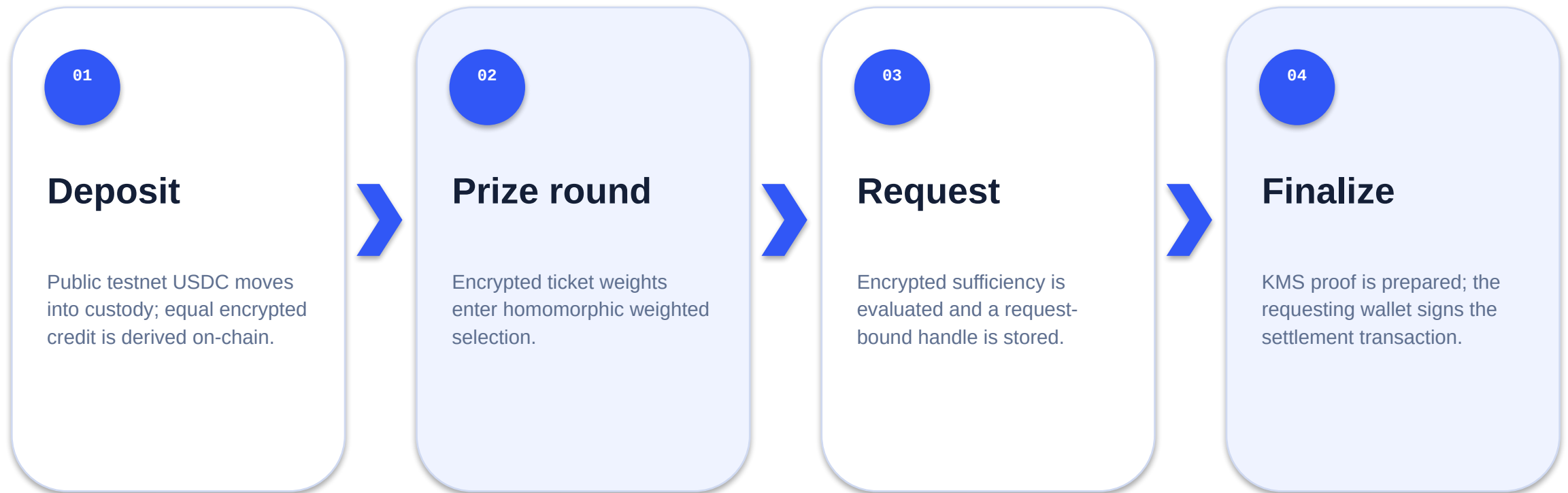
Clear wallet state

A disconnected user sees only Connect wallet. An address appears only after the provider returns the selected account.

Runtime assurance

Chain, bytecode, vault, and custody bindings are checked before protocol writes.

Four explicit steps. No hidden signing.



SECURITY CONTROLS

Solvency and settlement invariants are enforced in code

01

Asset-bound credit

Encrypted deposit credit is derived from the transferred public amount.

02

Reserved prizes

Each draw consumes available yield so assets cannot fund repeated awards.

03

Complete liabilities

Compounded prizes increase aggregate accounted obligations.

04

Anchored requests

The encrypted withdrawal handle is stored and bound to the requester.

05

Verifiable exit

KMS signatures are checked on-chain; stale requests can cancel after 24 hours.

VERIFIED EVIDENCE

A real 1 USDC Sepolia lifecycle—not a simulated receipt

Each transaction below links directly to its confirmed block-explorer record.

01

Deposit

[0x934340...8e2e](#)

Custody +1 USDC

02

Withdrawal request

[0x31ae42...7a6b](#)

Handle anchored

03

KMS finalization

[0xc160a1...3878](#)

Wallet restored

Final state: wallet 20 USDC · pool custody 0 · accounted principal 0

113 automated tests—and durable live infrastructure

113

PASSING TESTS

51 Foundry · 32 backend
2 client · 28 frontend

Vercel

React frontend

PRODUCTION UI

Render

Node relayer/indexer

PUBLIC API

Neon

PostgreSQL checkpoint

RESTART RECOVERY

Indexer checkpoint restored successfully after a live service restart

OPEN RESEARCH · VERIFIABLE EVIDENCE

Confidential savings, without blind trust.

Try the live app. Inspect the active Sepolia deployment. Review every invariant and test.

cipherpool-beta.vercel.app

github.com/Webghost01-NG/fhevm-pooltogether-security

ACTIVE POOL

0x105C5786
0b32a37F
3C7CF2AE
cF5a39Ab
bCA1d265

ETHEREUM SEPOLIA